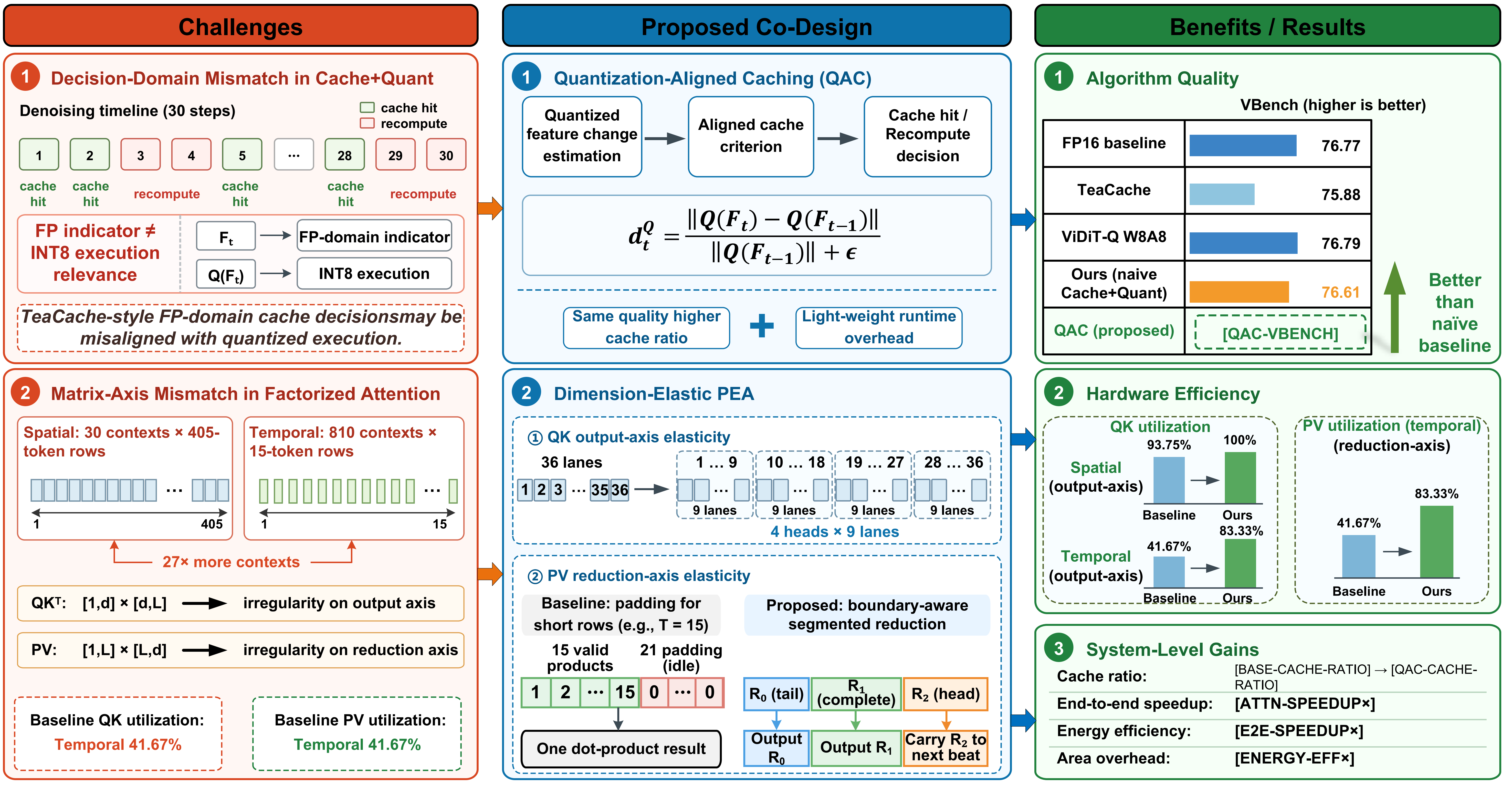
Challenges
1 Decision-Domain Mismatch in Cache+Quant
Denoising timeline (30 steps)
cache hit
recompute
1 2 3 4 5 ... 28 29 30
cache hit
cache hit
recompute
cache hit
cache hit
recompute
FP indicator ≠ INT8 execution relevance
F_t → FP-domain indicator
Q(F_t) → INT8 execution
TeaCache-style FP-domain cache decisionsmay be misaligned with quantized execution.
2 Matrix-Axis Mismatch in Factorized Attention
Spatial: 30 contexts × 405-token rows
1 405
Temporal: 810 contexts × 15-token rows
1 15
27× more contexts
QKᵀ: [1,d] × [d,L] irregularity on output axis
PV: [1,L] × [L,d] irregularity on reduction axis
Baseline QK utilization: Temporal 41.67%
Baseline PV utilization: Temporal 41.67%
Proposed Co-Design
1 Quantization-Aligned Caching (QAC)
Quantized feature change estimation
Aligned cache criterion
Cache hit / Recompute decision
$d_t^Q = \frac{\|Q(F_t) - Q(F_{t-1})\|}{\|Q(F_{t-1})\| + \epsilon}$
Same quality higher cache ratio
+
Light-weight runtime overhead
2 Dimension-Elastic PEA
① QK output-axis elasticity
36 lanes
1 2 3 ... 35 36
1 ... 9
10 ... 18
19 ... 27
28 ... 36
9 lanes
9 lanes
9 lanes
9 lanes
4 heads × 9 lanes
② PV reduction-axis elasticity
Baseline: padding for short rows (e.g., T = 15)
15 valid products
21 padding (idle)
1 2 ... 15 0 ... 0
One dot-product result
Proposed: boundary-aware segmented reduction
R₀ (tail)
R₁ (complete)
R₂ (head)
Output R₀
Output R₁
Carry R₂ to next beat
Benefits / Results
1 Algorithm Quality
VBench (higher is better)
FP16 baseline 76.77
TeaCache 75.88
ViDiT-Q W8A8 76.79
Ours (naive Cache+Quant) 76.61
QAC (proposed) [QAC-VBENCH]
Better than naïve baseline
2 Hardware Efficiency
QK utilization
Spatial (output-axis)
93.75%
100%
Baseline
Ours
Temporal (output-axis)
41.67%
83.33%
Baseline
Ours
PV utilization (temporal) (reduction-axis)
41.67%
83.33%
Baseline
Ours
3 System-Level Gains
Cache ratio: [BASE-CACHE-RATIO] → [QAC-CACHE-RATIO]
End-to-end speedup: [ATTN-SPEEDUP×]
Energy efficiency: [E2E-SPEEDUP×]
Area overhead: [ENERGY-EFF×]